

Curriculum Vitae – SHURUI LIN (林书睿)

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Education

Ph.D. Astronomy

University of Illinois Urbana-Champaign

August 2024 – Present

B.S. Physics

University of Science and Technology of China

August 2020 – July 2024

Awards and Achievements

2024 Honor Student, Yan Jici Talent Program in Physics

2022 Silver Medal, Exceptional Student Scholarship, USTC

2021 Gold Medal, Exceptional Student Scholarship, USTC

First Author Publications

1. **Shurui Lin**, Yin Li, Shy Genel, Francisco Villaescusa-Navarro, Biwei Dai, Wentao Luo, and Yang Wang.

One Latent to Fit Them All: A Unified Representation of Baryonic Feedback on Matter Distribution.
The Astrophysical Journal Letters, 996(2).

2. **Shurui Lin**, Xiangchong Li, Ji Li, Shengcao Cao, Xin Liu, and Yu-Xiong Wang.

D₄CNN×AnaCal: Physics-Informed Machine Learning for Accurate and Precise Weak Lensing Shear Estimation.

Accepted by The Astrophysical Journal, arXiv:2603.19046 (2026).

3. **Shurui Lin**, Xiangchong Li, Xin Liu

Slay the Shear: A Unified Statistical Framework for Weak Gravitational Lensing Shear Estimation.

Submitted The Astrophysical Journal, arXiv:2606.25596 (2026).

4. **Shurui Lin**, Francisco Villaescusa-Navarro, Jonah Rose, Paul Torrey, Arya Farahi, Kassidy E. Kollmann, Alex M. Garcia, Sandip Roy, Mark Vogelsberger, Nitya Kallivayalil, Yi-Fu Cai, Wentao Luo.

Can We Constrain Warm Dark Matter Masses with Individual Galaxies? The Astrophysical Journal, 970(2).

5. **Shurui Lin**, Tanda Li, Shude Mao, and Jim Fuller.

Asteroseismology with TESS: Emergence of Dipole Mode Suppression From Subgiants? The Astrophysical Journal, 980(2).

6. **S. Lin**, W. Luo, Y. F. Cai, Q. Guo, L. Wei, B. Wang, Q. Li, C. Su, and A. Rodriguez.

A ~ 300 pc-sized core of Milky Way dark matter halo constrained from the OGLE micro-lensing sky map. A&A, 693, L8 (2025).

Co-authored Publications

1. Qi Guo, Leyao Wei, Wentao Luo, **Shurui Lin**, Qinxun Li, Yi-Fu Cai, Di He, Qingqing Wang, and Ruoxi Yang. *Static Microlensing: Concept, Method and Candidates.*

2. Andrew Eberhardt, Elisa Ferreira, Wentao Luo, **Shurui Lin**, Yin Li. *Stochastic lensing of stars by ultralight dark matter halos.*

Presentation

2026 HSC-UNIONS-WFST collaboration meeting

Physics-Informed Machine Learning for Weak-Lensing Shear Estimation

2026 Tsinghua University

Slay the Shear: Physics-Informed Machine Learning for Weak-Lensing Shear Estimation

2026 Westlake University

Slay the Shear: Physics-Informed Machine Learning for Weak-Lensing Shear Estimation

2026 University of Science and Technology of China

Slay the Shear: Physics-Informed ML for WL Shear Estimation

2026 Shanghai Jiao Tong University

Slay the Shear: A Unified Statistical Framework for WL Shear Estimation

2023 Academic Conference of Astronomical Society of Anhui Province

A ~ 300 pc-sized core of Milky Way dark matter halo constrained from the OGLE micro-lensing sky map

<https://doi.org/10.1051/0004-6361/202452408>